

Statistical Models in Computational Biology

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Please submit your project with the filename `LastName(s)_Project10.pdf`.

Problem 24: Uniqueness of NMF solutions

(2 points)

Given a non-negative matrix factorization

$$V \approx WH \quad \text{with} \quad V \in \mathbb{R}_{\geq 0}^{M \times N}, W \in \mathbb{R}_{\geq 0}^{M \times K}, H \in \mathbb{R}_{\geq 0}^{K \times N},$$

are the factor matrices unique? If they are not, provide alternative factor matrices $\tilde{W} \neq W$ and $\tilde{H} \neq H$ such that

$$(1) \quad \tilde{W}\tilde{H} = WH \quad \text{and} \quad \tilde{W} \in \mathbb{R}_{\geq 0}^{M \times K}, \tilde{H} \in \mathbb{R}_{\geq 0}^{K \times N}.$$

If they are unique, provide a proof that eq.(1) necessarily implies $\tilde{W} = W$ and $\tilde{H} = H$.

Problem 25: NMF of spatial gene expression patterns

(4 points)

The file `DrosophilaExpressions.rda` contains a data matrix V derived from early stage embryos of the fruit fly *Drosophila melanogaster* and a function `imageBatchDisplay` for its visualization¹. Each column of V is an observation of a different gene in terms of its spatial expression pattern. Each row of V is a feature that represents a different location (pixel) in the embryo.

Install and load the package NMF from CRAN to perform the following analyses.

1. Display the first 16 observations using `imageBatchDisplay(V[,1:16])` as elliptical images that resemble the embryo. Compute a factorization $V \approx \hat{V} = WH$ with `rank=15`, `seed=123` and the default `method="brunet"`, which corresponds to multiplicative updates algorithm with generalized KL divergence. Report the generalized KL divergence² of the approximated data matrix $\hat{V}$ and display its first 16 columns as images. Then display all computed basis patterns, i.e. columns of W . Repeat these steps for a rank 10 factorization.

(1.5 points)

2. For the rank 10 factorization, display the expression pattern of the gene `Mkp3` approximately as a linear combination of the computed basis patterns. To this end, pass the corresponding coefficients as the argument `imgNames` to `imageBatchDisplay` in order to print the coefficients below the basis patterns. Then do the same for the gene `CG31909`.

How can you interpret these results - more specifically, what do these computed patterns and the coefficients represent?

(2.5 points)

¹the data and code are taken from the [Berkeley Drosophila Genome Project](#) with slight modifications

²The NMF package reports the KL divergence misleadingly as "residuals"

Problem 26: Implementing NMF from scratch**(4 points)**

Implement the NMF algorithm on your own in R and test it by computing a rank 10 factorization WH of the matrix V from `DrosophilaExpressions.rda`. To this end, set a seed for reproducibility and initialize the entries of W and H with random numbers drawn uniformly between 0 and 1. Then iteratively optimize the generalized KL divergence using the multiplicative update rules

$$H_{kj} \leftarrow H_{kj} \frac{\sum_i W_{ik} V_{ij} / (WH)_{ij}}{\sum_i W_{ik}} \quad \text{and} \quad W_{ik} \leftarrow W_{ik} \frac{\sum_j H_{kj} V_{ij} / (WH)_{ij}}{\sum_j H_{kj}}.$$

However, do not directly implement the entry-wise formulae, as this would be inefficient. Instead, use higher level operations in R. These include the matrix product $A \%*\% B$, transpose $t(A)$, `rowSums(A)`, `colSums(A)` and the elementwise product $A * B$. The operation $A * v$ multiplies each column of A elementwise with a vector v . Elementwise division is analogous.

Report the generalized KL divergence³ after 1000 iterations, where one iteration updates both matrices. Visualize the inferred basis patterns in W as in problem 28. Does your result look similar to the result from the package?

³when some entries of V are exactly zero, use the convention $0 \log(0) = 0$ to compute the generalized KL divergence